

# Non-Contact Physical Health Profiling from Human Body Silhouettes Using Body Shape Embeddings

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## Abstract

Non-communicable diseases associated with overweight and obesity are among the most pressing public health challenges globally and within the United Arab Emirates (UAE). According to national health estimates, nearly 68% of adults aged 18 to 69 years in the UAE are classified as overweight or obese, with approximately 27 to 28% living with obesity alone and combined risk factors for obesity continuing to increase<sup>1</sup>. These conditions contribute substantially to diabetes, cardiovascular disease, and reduced quality of life, while traditional health screening systems still depend on clinical visits and specialized equipment. We aim to propose a non-intrusive, vision-based physical health assessment framework that leverages subtle visual cues such as human silhouettes and posture extracted from multiview images to estimate clinically relevant indicators of physical health, including body mass index (BMI) and body fat percentage. Our architecture integrates a foundational segmentation backbone for accurate silhouette generation, a contrastive shape representation module aligned with anthropological reference data, and a regression head grounded in medical knowledge to produce interpretable physical health estimates. By collaborating with healthcare professionals in the UAE, we ensure that our mapping between visual patterns and anthropometric phenotypes is medically valid and can assist in early detection of physical health risks, community health monitoring, and large-scale screening without traditional medical infrastructure. This research also aligns with multiple United Nations Sustainable Development Goals (SDGs), including SDG 3, SDG 10 and SDG 9. Our aim is to demonstrate how non-intrusive computer vision can provide actionable physical health insights, paving the way for scalable, respectful, and context-aware health screening for diverse populations.

<sup>1</sup><https://data.worldobesity.org/country/united-arab-emirates-225/>

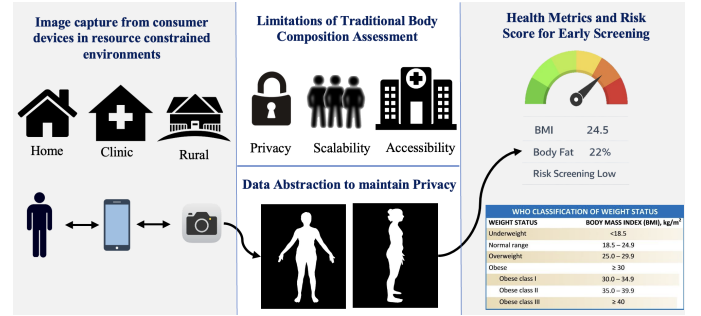


Figure 1: Overview of a privacy-preserving, non-intrusive physical health screening framework that uses multiview human silhouettes captured from consumer devices to estimate body composition metrics (BMI, body fat) and enable early risk assessment without specialized medical equipment.

## 1 Introduction

Body Mass Index (BMI), Body Adiposity Index (BAI), and Body Fat Percentage (BF) are widely used indicators for understanding body composition and assessing health risks related to undernutrition, excess weight, and obesity, particularly in large-scale population screening scenarios [Nana *et al.*, 2022; Xie *et al.*, 2015]. Although conceptually simple, accurate estimation of these indices generally relies on direct anthropometric measurements and access to clinical infrastructure or specialized equipment such as Dual Energy X-Ray Absorptiometry (DXA) or hydrostatic weighing [Nana *et al.*, 2022; Nascimento *et al.*, 2025; Bosy-Westphal *et al.*, 2018; Karlsson *et al.*, 2015; Ellis, 2000]. Such procedures are costly, time-consuming, and impractical for frequent or large-scale screening, especially in resource-constrained or community settings. This limitation is particularly pressing in regions like the United Arab Emirates, where a substantial proportion of the adult population is affected by overweight and obesity, increasing the burden of non-communicable diseases [World Obesity Federation, 2023; World Health Organization, 2022; Radwan *et al.*, 2018]. These challenges motivate the need for non-intrusive, scalable, and privacy-preserving alternatives that can enable early physical health assessment without requiring explicit medical equipment, clinical supervision, or invasive measurements.

Recent advances in computer vision and representation

learning have enabled the use of image-based methods for estimating human body characteristics from visual data [Ruiz *et al.*, 2022; Yan *et al.*, 2021]. By exploiting geometric cues, shape priors, and learned visual patterns, such approaches offer the possibility of non-intrusive physical health assessment, reducing reliance on explicit anthropometric measurements [Zhu *et al.*, 2023; Bogo *et al.*, 2016]. However, the growing adoption of connected medical devices and digital health systems has simultaneously heightened concerns around data security, privacy, and ethical use [Organization and others, 2021; Saxena and Mittal, 2022]. Methods that operate directly on raw RGB images inherently capture identifiable biometric information, raising risks related to identity exposure and misuse, especially in healthcare-oriented deployments [Nana *et al.*, 2022; Ruiz *et al.*, 2022]. These challenges pose significant barriers to real-world adoption, where clinical validity, interpretability, and regulatory compliance are critical. Consequently, the development of vision-based health assessment systems necessitates close collaboration with medical practitioners and healthcare professionals to ensure that learned visual patterns are grounded in established anthropometric and clinical knowledge.

## 1.1 Problem Statement

This project addresses the problem of screening and early risk assessment of body composition-related health conditions using visual data, with a focus on indicators such as Body Mass Index (BMI) and Body Adiposity Index (BAI). The objective is to design, develop, and validate a non-intrusive, vision-based framework capable of estimating these indicators in a privacy-preserving manner, even in resource-constrained settings where access to clinical infrastructure and standardized anthropometric measurements is limited or culturally sensitive. The proposed framework operates on silhouette representations derived from monocular smartphone images, abstracting human body shape while deliberately removing identifiable visual features.

A key challenge lies in ensuring robustness under unconstrained consumer capture conditions. To this end, the framework explicitly accounts for segmentation instability, viewpoint variability, and clothing-induced distortions, which commonly affect silhouette quality in real-world scenarios [Jin *et al.*, 2020; Lan *et al.*, 2022]. Additionally, the system incorporates uncertainty-aware estimation mechanisms to identify predictions that may be unreliable, thereby supporting cautious interpretation and referral rather than automated clinical decision-making [Begoli *et al.*, 2019]. The system is intended to function as an assistive screening and risk awareness tool, not as a diagnostic substitute. Following system development, extensive experimental evaluation will be conducted to assess estimation accuracy, robustness across capture conditions, and limitations arising from proxy supervision. The long-term goal is to enable accessible, scalable, and privacy-preserving physical health screening technologies that can complement existing healthcare workflows, particularly in environments where conventional body composition assessment procedures are impractical or unavailable, thereby further bridging the gap between computer vision research and real world public health screening.

## 1.2 Strategy

In this work, we investigate a silhouette-driven pipeline for estimating body composition indicators from monocular images captured using smartphones. By representing the human body through silhouettes, the proposed approach preserves coarse geometric and shape information while suppressing identity-sensitive visual features, thereby supporting privacy-preserving health screening [Xie *et al.*, 2015; Gardner *et al.*, 1998]. The primary objective of this study is to evaluate the feasibility of silhouette-based representations for body composition estimation in a manner that preserves privacy, reduces computational overhead, and enables scalable deployment. While silhouettes offer strong privacy guarantees, they also significantly compound the inference problem. A silhouette corresponds to a 2D projection of a 3D human body under unconstrained imaging conditions, including variations in camera distance, viewpoint, background, lighting, and clothing. Minor segmentation artifacts, such as holes, jagged edges, noise, or disconnected components can induce large perturbations in perceived widths and areas, which may propagate into downstream predictions [Hofmann *et al.*, 2025]. This sensitivity is particularly pronounced in consumer capture settings, where acquisition conditions can only be partially standardized (e.g., approximate camera height or subject distance using simple visual cues) but cannot be fully controlled [Sengupta *et al.*, 2021; Nana *et al.*, 2022; Yan *et al.*, 2021].

To address these challenges, the proposed framework employs an end-to-end processing pipeline that converts RGB smartphone images into standardized binary silhouettes and estimates body composition metrics from single-view or dual-view inputs. The preprocessing stage utilizes a segmentation module based on the Segment Anything Model (SAM) [Kirillov *et al.*, 2023], combined with attentive prompting and anatomical plausibility checks, to generate fixed-resolution  $640 \times 480$  silhouette masks from variable-resolution inputs. Post-processing steps are applied to mitigate common segmentation failures and enforce structural consistency prior to feature extraction. The system adopts a two-branch architecture, wherein front-view and side-view silhouettes are processed independently to extract shape embeddings that capture complementary geometric cues. These embeddings are subsequently synchronized using a cross-modal contrastive learning objective, designed to align related silhouette pairs while reducing false negatives through precise sample association [Zolfaghari *et al.*, 2021]. This design enables robust representation learning under viewpoint variation and partial silhouette degradation. Additionally, the architecture draws inspiration from BMNet-style approaches that estimate three-dimensional body measurements from silhouette inputs by learning mappings aligned with anthropometric measurements [Ruiz *et al.*, 2022].

Due to the lack of large-scale clinical ground truth for body fat percentage (BF%) within the current data collection framework, the system initially employs Body Adiposity Index (BAI) as a surrogate supervision signal. Consequently, predicted BF% values should be interpreted as approximate estimates rather than clinically definitive measurements [López *et al.*, 2012]. A key objective of this project is

to transition from proxy supervision to direct BF% calibration by constructing a dataset of smartphone captures paired with reference measurements obtained via multi-frequency bio-electrical impedance analysis (BIA). Using an 8-point electrode system (hands and feet), BIA estimates body composition by measuring tissue-dependent impedance responses to low-amplitude electrical currents, enabling detailed assessment of fat mass, lean mass, and related parameters [Buckinx *et al.*, 2015; Velázquez-Alva *et al.*, 2022]. This dataset will support clinically meaningful evaluation, calibration, and validation of the proposed vision-based screening framework.

### 1.3 Expected Outcomes and Impact

This project is expected to deliver a non-intrusive, privacy-preserving vision-based framework for physical health screening, developed through close collaboration between computer vision researchers and healthcare professionals within the UAE ecosystem. A key outcome will be a validated pipeline that integrates silhouette-based visual representations with expert-curated anthropometric and clinical knowledge, enabling reliable estimation of body composition indicators suitable for early risk awareness.

From a research perspective, the project will advance methodological understanding of shape-based representation learning under privacy constraints, particularly in unconstrained consumer capture settings. The study will also contribute curated datasets, collected under appropriate ethical approvals linking smartphone-captured visual data with clinically grounded body composition measurements, supporting reproducibility and further research in non-intrusive health assessment.

### 1.4 Alignment with Sustainable Development Goals (SDGs)

This work addresses the real-world challenge of enabling accessible and preventive screening for body composition-related health risks in settings where conventional clinical measurements are impractical, unavailable, or culturally sensitive. The problem formulation and system design are informed through consultative collaboration with healthcare professionals, whose advisory input guides the clinical relevance of body composition indicators, practical constraints of existing workflows, and ethical considerations surrounding privacy and non-intrusive data collection. The proposed framework directly aligns with SDG 3 (Good Health and Well-Being), particularly Target 3.4, by supporting early risk identification for non-communicable diseases associated with obesity and malnutrition through contactless estimation of body composition indicators such as BMI and adiposity-related measures. It also contributes to Target 3.8 by lowering barriers to basic health screening through the use of consumer-grade devices, reducing reliance on specialized clinical equipment, and enabling population-scale health monitoring. In addition, the work aligns with SDG 10 (Reduced Inequalities), specifically Target 10.2, by promoting inclusive access to preventive health assessment in resource-limited, geographically constrained, or culturally sensitive contexts. By minimizing dependence on costly infrastructure and standardized clinical environments, the proposed ap-

proach aims to mitigate disparities in access to early health screening. The project also aligns with SDG 9 (Industry, Innovation, and Infrastructure), particularly Target 9.1, by leveraging scalable, low-cost digital infrastructure and consumer graded sensing technologies to support resilient and inclusive health screening systems in resource limited settings.

## 2 Related Work

Accurate assessment of body composition has traditionally relied on clinical and imaging-based techniques, including dual-energy X-ray absorptiometry (DXA), air-displacement plethysmography (e.g., BodyPod), magnetic resonance imaging (MRI), and computed tomography (CT). These methods provide direct estimates of fat mass, lean mass, and regional adiposity distribution, and serve as reference standards in clinical research [Heymsfield, 2005; Fields *et al.*, 2002]. In particular, whole-body MRI has enabled large-scale quantification of subcutaneous and visceral adipose tissue and has supported the development of automated segmentation pipelines for population studies [Thomas *et al.*, 2013; Addeman *et al.*, 2015; Yan *et al.*, 2021; Küstner *et al.*, 2020]. Despite their accuracy, such techniques require specialized equipment and controlled settings, limiting their scalability and accessibility. Prior studies have demonstrated that silhouettes retain informative morphological cues correlated with anthropometric measurements and health-related traits, particularly at the population level [Yan *et al.*, 2021; Ruiz *et al.*, 2022]. These findings motivate the use of shape-only cues for health assessment under privacy constraints. Among silhouette-based methods, multi-view approaches have shown particular promise. The BodyM/BMNet paradigm leverages paired front and side silhouettes together with anthropometric supervision to directly regress body measurements, avoiding explicit three-dimensional mesh reconstruction [Ruiz *et al.*, 2022]. This design reduces model complexity and supervision requirements compared to full body reconstruction pipelines, while still exploiting complementary information across orthogonal views. Concurrent benchmarking efforts further highlight that silhouette quality, standardization, and viewpoint consistency play a critical role in downstream estimation accuracy [Gardner *et al.*, 1998].

An alternative line of work focuses on three-dimensional body reconstruction, often using parametric models such as SMPL-like meshes, from which anthropometric measurements are derived [Kim *et al.*, 2022]. While these approaches provide detailed geometric representations and improved interpretability, they introduce additional sources of error related to pose-shape disentanglement and model mismatch, and typically require stronger priors or denser supervision. Finally, multi-view representation learning with contrastive objectives has emerged as an effective mechanism for aligning embeddings across different views of the same subject without requiring additional labels. Contrastive learning naturally enforces consistency between front and side silhouettes while preserving discrimination across individuals, making it well suited for privacy-preserving body shape analysis [Zolfaghari *et al.*, 2021]. Building on these insights, the

Table 1: WHO Body Mass Index (BMI) Classification for Adults

| BMI Range (kg/m <sup>2</sup> ) | Category      | Subcategory |
|--------------------------------|---------------|-------------|
| < 18.5                         | Underweight   | –           |
| 18.5 – 24.9                    | Normal weight | –           |
| 25.0 – 29.9                    | Overweight    | –           |
| 30.0 – 34.9                    | Obese         | Class I     |
| 35.0 – 39.9                    | Obese         | Class II    |
| ≥ 40.0                         | Obese         | Class III   |

present work adopts a multi-view silhouette-based formulation for non-intrusive body composition screening, with predicted Body Mass Index (BMI) values interpreted according to established clinical standards. In particular, BMI-based health categories follow the World Health Organization (WHO) adult classification guidelines [World Health Organization, 2024], as summarized in Table 1.

### 3 System Description

#### 3.1 Dataset

We use *BodyM* [Ruiz *et al.*, 2022] as the primary reference dataset, as it provides real-world silhouette images paired with anthropometric measurements. *BodyM* contains 8,978 frontal and lateral silhouette images corresponding to 2,505 individuals, along with height, weight, and 14 additional body measurements. The silhouettes are generated via semantic segmentation applied to real RGB photographs, thereby preserving realistic segmentation artifacts that are typically absent in purely synthetic silhouette datasets. The data acquisition setup is controlled but representative of consumer capture conditions: images are captured indoors under uniform lighting, subjects adopt an A-pose while wearing tight-fitting clothing, and the reported camera-to-subject distance ranges from 5.5 to 6.5 feet. The dataset is partitioned at the subject level, with 2,018/87/400 individuals assigned to the Train, Test-A, and Test-B splits, respectively, corresponding to 6,134/1,684/1,160 silhouette images.

**Related Datasets.** *CAESAR* [Robinette *et al.*, 1999] provides large-scale 3D body scans with associated anthropometric measurements; however, silhouettes or images must be synthetically rendered from the scans. *BodyFit* [Yan *et al.*, 2021], which builds upon such 3D resources, includes over 4,000 high-resolution body scans with computed measurements and simulated silhouettes, along with a smaller subset of 194 subjects captured using real photographs and tape-based measurements. *BodyFit* assumes a fixed camera distance to resolve scale, which limits its applicability in unconstrained consumer capture scenarios.

#### 3.2 Method Overview

We estimate body health metrics, including BMI and BF%, from binary silhouette representations captured using a smartphone. The proposed pipeline consists of three main stages: (i) extracting front and side-view silhouettes from RGB images, (ii) standardizing silhouettes to a fixed spatial resolution of  $640 \times 480$ , and (iii) predicting BMI and a BF%-proxy us-

ing a dual-branch CNN trained with cross-view contrastive alignment and supervised regression.

All images are captured in an unsupervised environment using an iPhone at a fixed camera height of 1.25 m, with the subject positioned at a distance in the range [1.8 m, 2.25 m].

#### Silhouette Extraction and Standardization

Let  $I$  denote an RGB image captured by a smartphone. A smartphone image processing pipeline is used to extract a binary human silhouette mask  $M$  from  $I$ , where foreground pixels take the value of 255 and background pixels are 0:

$$M \in \{0, 255\}^{H \times W}.$$

To enable consistent downstream processing, each silhouette is standardized using a deterministic resizing and centering operation:

$$M_{\text{std}} = \mathcal{R}(M),$$

where

$$\mathcal{R}(M) = \text{center}(\text{scale}(\text{crop-to-bbox}(M))).$$

such that scaling preserves aspect ratio and centering is performed via padding.

This operation ensures scale normalization and spatial alignment across samples while preserving body shape proportions. Separate refinement procedures are applied to front-view and side-view silhouettes to mitigate view-specific segmentation artifacts and shape distortions commonly observed in consumer capture scenarios.

#### Dual-Branch Representation Learning

Separate refinement procedures are applied to front-view and side-view silhouettes to mitigate view-specific segmentation artifacts and shape distortions commonly observed in consumer capture scenarios. Let  $x_f, x_s \in \{0, 1\}^{640 \times 480}$  denote the standardized binary silhouettes corresponding to the front and side views, respectively. The network employs a dual-branch architecture with separate encoders for each view:

$$h_f = E_f(x_f), \quad h_s = E_s(x_s),$$

where each encoder consists of a lightweight convolutional stack followed by global average pooling and fully connected layers. The resulting embeddings are concatenated to form a fused representation:

$$\mathbf{h} = [\mathbf{h}_f \mid \mathbf{h}_s].$$

This design enables the model to capture complementary geometric cues from orthogonal viewpoints while remaining computationally efficient.

#### Prediction Heads

**Measurement head.** The measurement head predicts anthropometric quantities, including BMI, and is defined as

$$\hat{y}_{\text{meas}} = W_m \mathbf{h} + b_m.$$

**Body fat head.** The body fat head produces a sigmoid-bounded output in the range  $[0, 1]$ , which is later scaled to represent percentage values:

$$\hat{y}_{\text{bf}} = \sigma(W_b \mathbf{h} + b_b).$$

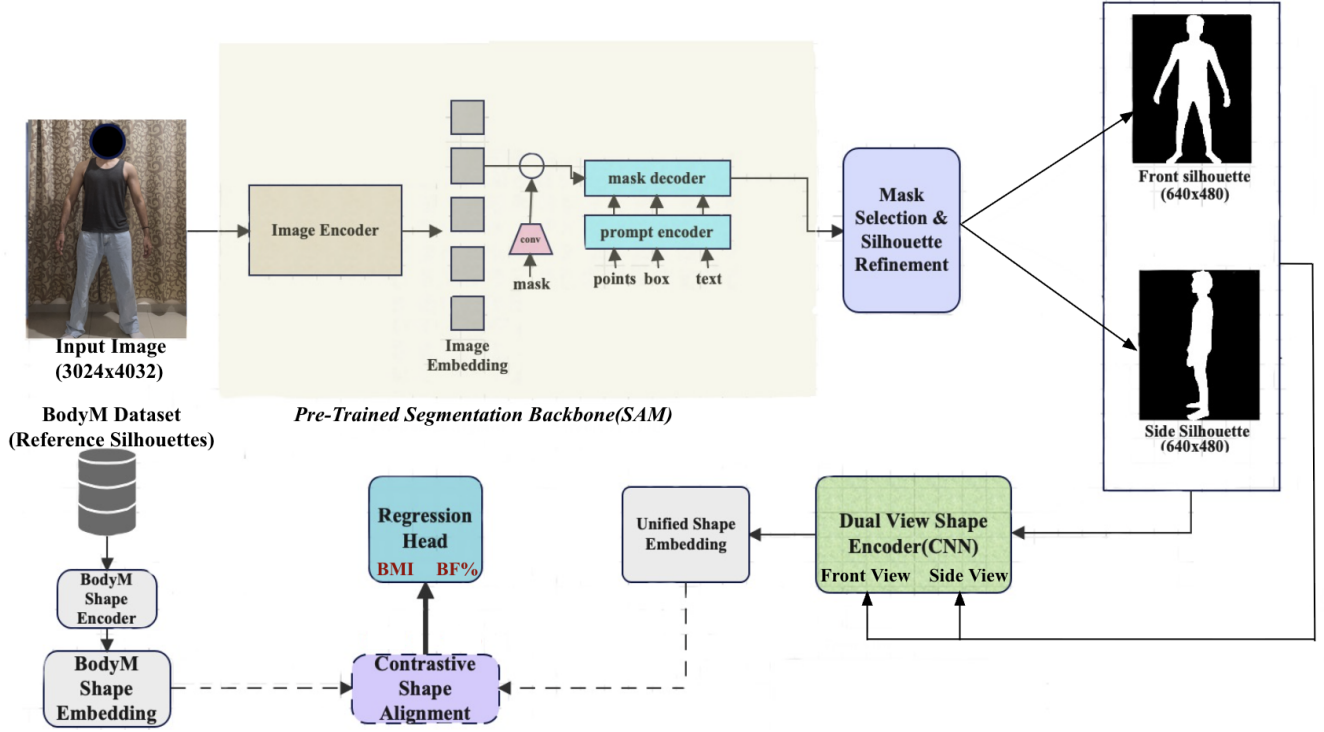


Figure 2: Overview of the proposed privacy-preserving architecture for non-intrusive body composition screening, which extracts front and side silhouettes using a SAM-based segmentation pipeline and estimates BMI and body fat-related measures via dual-view shape encoding and contrastive alignment.

### 3.3 Training Objectives

#### Cross-View Contrastive Alignment

To enforce consistency between front- and side-view representations, the model employs an InfoNCE (NT-Xent) style contrastive loss [Oord *et al.*, 2018; Chen *et al.*, 2020; Zolfaghari *et al.*, 2021]. Let the normalized projected embeddings be

$$\mathbf{z}_f = \frac{g(\mathbf{h}_f)}{\|g(\mathbf{h}_f)\|}, \quad \mathbf{z}_s = \frac{g(\mathbf{h}_s)}{\|g(\mathbf{h}_s)\|},$$

where  $g(\cdot)$  denotes a projection head and  $\|\cdot\|$  is the  $\ell_2$  norm.

For a batch of size  $N$ , with positive pairs  $(i, i)$  and negative pairs  $(i, j)$  for  $j \neq i$ , the contrastive loss is defined as

$$\mathcal{L}_{\text{con}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\mathbf{z}_{f,i}^\top \mathbf{z}_{s,i} / \tau)}{\sum_{j=1}^N \exp(\mathbf{z}_{f,i}^\top \mathbf{z}_{s,j} / \tau)},$$

where  $\tau$  is a temperature scaling parameter.

#### Regression Supervision

Supervised learning is applied using  $\ell_1$  (MAE) losses for both anthropometric measurements and body fat estimation:

$$\mathcal{L}_{\text{reg}} = \|\hat{y}_{\text{meas}} - y_{\text{meas}}\|_1 + \|\hat{y}_{\text{bf}} - y_{\text{bf}}\|_1.$$

#### Total Loss

The overall training objective is a weighted combination of contrastive alignment and regression supervision:

$$\mathcal{L} = \mathcal{L}_{\text{con}} + \lambda \mathcal{L}_{\text{reg}},$$

where  $\lambda$  balances representation alignment and prediction accuracy.

#### Labels and Targets

Body Mass Index (BMI), defined as body weight divided by squared height [Keys *et al.*, 2014], is used as a primary anthropometric indicator.

**BMI label.** Given height  $h$  (in meters) and weight  $w$  (in kilograms), BMI is computed as

$$\text{BMI} = \frac{w}{h^2}.$$

**Body fat proxy.** Due to the limited availability of large-scale clinical body fat percentage (BF%) ground truth, the Body Adiposity Index (BAI) [Bergman *et al.*, 2011] is used as a proxy target:

$$\text{BAI} = \frac{\text{hip}_{\text{cm}}}{h_m^{1.5}} - 18.$$

During training, BAI values are scaled and clipped to obtain a sigmoid-compatible target:

$$y_{\text{bf}} = \text{clip}\left(\frac{\text{BAI}}{100}, 0, 1\right).$$

It is acknowledged that BAI may exhibit systematic biases due to its reliance on limited anthropometric measurements, with reduced accuracy reported across sex, ethnicity, and body fat distribution patterns [Geliebter *et al.*, 2013].



Table 2: Silhouette overlap metrics under the privacy-preserving evaluation protocol.

| Setting              | IoU $\uparrow$ | Dice $\uparrow$ |
|----------------------|----------------|-----------------|
| Post pixel alignment | 0.9015         | 0.9482          |

## 4 Experimental Setup and Preliminary Analysis

### 4.1 Silhouette Quality Evaluation

We evaluate the quality of the extracted silhouettes using spatial overlap metrics between a predicted silhouette and a reference silhouette. Since facial pixels are intentionally excluded as part of the privacy-preserving design, overlap evaluation is performed after removing the top portion of each mask. For a binary mask  $M$ , we define a modified mask  $M'$  by eliminating the top 25% of rows corresponding to the head and face region. This preprocessing is motivated by two factors. First, the available reference data include only two real-subject image pairs, and the IoU and Dice scores are therefore computed on this limited subset. Second, there is a general scarcity of publicly available datasets containing full-body human images captured in neutral poses with ground-truth silhouettes, particularly under privacy-preserving constraints. Silhouette overlap is evaluated using Intersection-over-Union (IoU) and the Dice similarity coefficient, both of which are standard metrics in medical image segmentation and shape analysis [Dice, 1945; Rezatofighi *et al.*, 2019]:

$$\text{IoU}(P, R) = \frac{|P \cap R|}{|P \cup R|}, \quad \text{Dice}(P, R) = \frac{2|P \cap R|}{|P| + |R|}$$

Results summarized in Table 2 suggest that silhouettes retain body contour structure with high spatial overlap relative to the reference silhouettes, supporting the suitability of silhouette-based representations as robust inputs for downstream body composition estimation.

### 4.2 Training Dynamics

To analyze optimization behavior and training stability, we monitor multiple quantities across training epochs, including:

- training and validation total loss (raw and smoothed),
- decomposition of contrastive and regression loss components,
- validation BMI mean absolute error (MAE), and
- validation BF% MAE in proxy-target space.

Across all experimental runs, both contrastive and regression losses decrease steadily, while validation MAE curves exhibit higher variance during early epochs followed by stabilization in later stages. This trend indicates stable optimization under the current supervision regime and suggests that the contrastive alignment objective does not impede convergence of the regression task. Figures 3a–3d illustrate the training and validation loss curves, loss decomposition, and

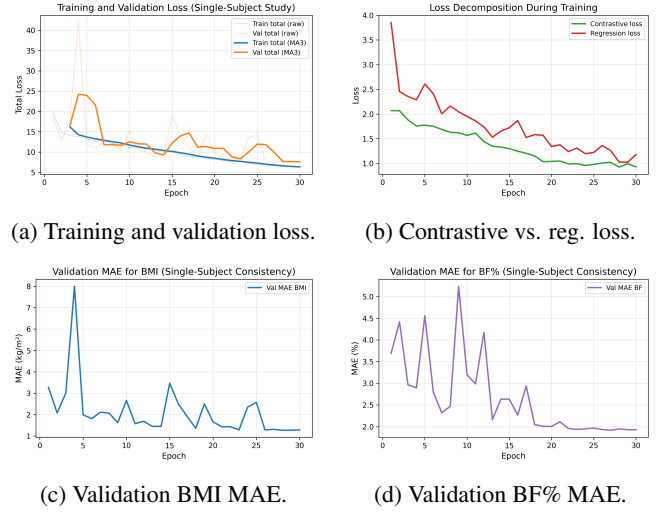


Figure 3: Training dynamics and validation performance of the proposed model: (a) training and validation loss, (b) loss decomposition, (c) BMI MAE, and (d) BF% MAE across epochs.

validation MAE trends. Quantitative results highlighting the impact of auxiliary metadata are reported in Table 3.

$$\text{MAE}(y, \hat{y}) = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

To further contextualize error magnitudes, we also report threshold-based true-positive rates defined as:

$$\text{TP@}\delta = 100 \cdot \frac{1}{N} \sum_{i=1}^N \mathbf{1}(|y_i - \hat{y}_i| \leq \delta), \quad \delta \in \{2.0, 1.0, 0.5\}$$

### 4.3 Results Analysis

Table 5 compares BMI prediction accuracy under single-view and multi-view configurations. The results indicate that incorporating a complementary viewpoint leads to a noticeable reduction in BMI prediction error on the evaluated samples, highlighting the benefit of multi-view shape information even under limited supervision. This improvement suggests that front and side silhouettes capture complementary geometric cues that are not fully observable from a single view.

Table 4 further analyzes the impact of contrastive alignment on both BMI and BF% estimation. Models trained with contrastive learning exhibit lower error metrics (MAE and RMSE) and higher correlation measures compared to their non-contrastive counterparts. These results indicate that contrastive alignment improves cross-view representation consistency, which in turn supports more stable and accurate regression of body composition indicators, particularly under constrained data conditions.

## 5 Challenges, Risks & Ethical Considerations

Data variability and bias pose risks, as consumer-captured images can exhibit variations in pose, clothing, lighting, and camera geometry, potentially impacting silhouette quality

Table 3: Impact of auxiliary metadata on estimation accuracy.

| Mode                         | TP@2 $\uparrow$ | TP@1 $\uparrow$ | TP@0.5 $\uparrow$ | BMI MAE $\downarrow$ | BAI MAE $\downarrow$ | Hip <sub>cm</sub> MAE $\downarrow$ |
|------------------------------|-----------------|-----------------|-------------------|----------------------|----------------------|------------------------------------|
| Single-View                  | 0.00            | 0.00            | 0.00              | 22.752               | 25.172               | 99.466                             |
| Multi-View                   | 0.00            | 0.00            | 0.00              | 22.873               | 25.298               | 99.587                             |
| Multi-View + Height          | 35.00           | 10.00           | 7.50              | 3.035                | 2.725                | 5.995                              |
| Multi-View + Weight          | 62.50           | 30.00           | 20.00             | 1.728                | 3.424                | 3.150                              |
| Multi-View + Height + Weight | 100.00          | 95.00           | 85.00             | 0.00                 | 1.251                | 2.472                              |

Table 4: Effect of contrastive learning on BMI and BF% estimation performance.

| Setting        | BMI MAE $\downarrow$ | BMI RMSE $\downarrow$ | BMI $R^2\uparrow$ | BMI $r\uparrow$ | BF% MAE $\downarrow$ | BF% RMSE $\downarrow$ | BF% $R^2\uparrow$ | BF% $r\uparrow$ |
|----------------|----------------------|-----------------------|-------------------|-----------------|----------------------|-----------------------|-------------------|-----------------|
| Contrastive    | 1.053                | 1.767                 | 0.931             | 0.969           | 1.940                | 2.428                 | 0.877             | 0.937           |
| No-Contrastive | 2.100                | 3.231                 | 0.770             | 0.908           | 4.449                | 6.929                 | -0.004            | 0.719           |

Table 5: Comparison of BMI and BF% prediction errors under single-view and multi-view settings

| Mode        | BMI (G.T) | BMI (Pred) | BMI MAE |
|-------------|-----------|------------|---------|
| Single-View | 21.46     | 22.75      | 1.29    |
| Multi-View  | 21.46     | 21.17      | 0.29    |

and downstream predictions. Limited availability of clinically validated ground-truth data for body fat percentage further constrains supervision and model calibration. These risks will be mitigated by incorporating robust preprocessing and data augmentation, uncertainty-aware estimation, gradual calibration using clinically measured data.

All human-related data used in this project will be collected with appropriate informed consent or sourced from publicly available datasets that explicitly permit research use. Where required, prior approval will be obtained from relevant Institutional Review Boards (IRBs) or ethics committees before initiating any data collection involving human participants. To protect participant privacy, all data will be anonymized prior to analysis. Raw images containing identifiable facial features or personal identifiers will not be stored or processed; instead, all visual data will be converted into silhouette-based representations before any modeling or experimentation. The project will be conducted in close collaboration with healthcare professionals to ensure that model development, evaluation, and interpretation adhere to established ethical and clinical standards.

## 6 Conclusion

We presented a privacy-preserving silhouette-based framework for estimating body health metrics, including BMI and BF%, from smartphone images. The system integrates SAM-based silhouette extraction with anatomy-aware refinements and a dual-branch contrastive model that processes frontal and lateral views, with support for single-view inference. Preliminary experiments demonstrate that the extracted silhouettes retain sufficient structural fidelity under a privacy-preserving evaluation protocol and that training remains stable under the proposed objectives. Ablation studies reveal

an inherent identifiability constraint: silhouettes alone provide limited predictive accuracy, while incorporating height and weight substantially improves performance. A single-subject analysis indicates accurate BMI estimation, whereas BF% estimates remain more variable, highlighting the limitations of proxy supervision. Future work will focus on expanding datasets with clinically measured BF% ground truth and introducing a WHO-aligned interpretability layer to map predictions to standardized health risk categories, improving transparency and translational relevance.

## References

- [Addeman *et al.*, 2015] Bryan T Addeman, Shelby Kutty, Thomas G Perkins, Abraam S Soliman, Curtis N Wiens, Colin M McCurdy, Melanie D Beaton, Robert A Hegele, and Charles A McKenzie. Validation of volumetric and single-slice mri adipose analysis using a novel fully automated segmentation method. *Journal of Magnetic Resonance Imaging*, 41(1):233–241, 2015.
- [Begoli *et al.*, 2019] Edmon Begoli, Tanmoy Bhattacharya, and Dimitri Kusnezov. The need for uncertainty quantification in machine-assisted medical decision making. *Nature Machine Intelligence*, 1(1):20–23, 2019.
- [Bergman *et al.*, 2011] Richard N Bergman, Darko Stefanovski, Thomas A Buchanan, Anne E Sumner, James C Reynolds, Nancy G Sebring, Anny H Xiang, and Richard M Watanabe. A better index of body adiposity. *Obesity*, 19(5):1083–1089, 2011.
- [Bogo *et al.*, 2016] Federica Bogo, Angjoo Kanazawa, Christoph Lassner, Peter Gehler, Javier Romero, and Michael J Black. Keep it smpl: Automatic estimation of 3d human pose and shape from a single image. In *European conference on computer vision*, pages 561–578. Springer, 2016.
- [Bosy-Westphal *et al.*, 2018] Anja Bosy-Westphal, Wiebke Braun, Corinna Geisler, Kristina Norman, and Manfred J Müller. Body composition and cardiometabolic health: the need for novel concepts. *European journal of clinical nutrition*, 72(5):638–644, 2018.

- [Buckinx *et al.*, 2015] Fanny Buckinx, Jean-Yves Reginster, Nadia Dardenne, Jean-Louis Croisier, Jean-François Kaux, Charlotte Beaudart, Justine Slomian, and Olivier Bruyère. Concordance between muscle mass assessed by bioelectrical impedance analysis and by dual energy x-ray absorptiometry: a cross-sectional study. *BMC musculoskeletal disorders*, 16(1):60, 2015.
- [Chen *et al.*, 2020] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *International conference on machine learning*, pages 1597–1607. PmLR, 2020.
- [Dice, 1945] Lee R Dice. Measures of the amount of ecological association between species. *Ecology*, 26(3):297–302, 1945.
- [Ellis, 2000] Kenneth J Ellis. Human body composition: in vivo methods. *Physiological reviews*, 80(2):649–680, 2000.
- [Fields *et al.*, 2002] David A Fields, Michael I Goran, and Megan A McCrory. Body-composition assessment via air-displacement plethysmography in adults and children: a review. *The American journal of clinical nutrition*, 75(3):453–467, 2002.
- [Gardner *et al.*, 1998] Rick M Gardner, Brenda N Friedman, and Natalie A Jackson. Methodological concerns when using silhouettes to measure body image. *Perceptual and motor skills*, 86(2):387–395, 1998.
- [Geliebter *et al.*, 2013] Allan Geliebter, Deniz Atalayer, Louis Flanchbaum, and Charlisa D Gibson. Comparison of body adiposity index (bai) and bmi with estimations of % body fat in clinically severe obese women. *Obesity*, 21(3):493–498, 2013.
- [Heymsfield, 2005] Steven Heymsfield. *Human body composition*, volume 918. Human kinetics, 2005.
- [Hofmann *et al.*, 2025] Felix O Hofmann, Christian Heiliger, Tengis Tschaidse, Stefanie Jarmusch, Liv A Auhage, Ughur Aghamaliyev, Alena B Gesenhues, Tobias S Schiergens, Hanno Niess, Matthias Ilmer, et al. Validation of body composition parameters extracted via deep learning-based segmentation from routine computed tomographies. *Scientific reports*, 15(1):11909, 2025.
- [Jin *et al.*, 2020] Sheng Jin, Lumin Xu, Jin Xu, Can Wang, Wentao Liu, Chen Qian, Wanli Ouyang, and Ping Luo. Whole-body human pose estimation in the wild. In *European Conference on Computer Vision*, pages 196–214. Springer, 2020.
- [Karlsson *et al.*, 2015] Anette Karlsson, Johannes Rosander, Thobias Romu, Joakim Tallberg, Anders Grönqvist, Magnus Borga, and Olof Dahlqvist Leinhard. Automatic and quantitative assessment of regional muscle volume by multi-atlas segmentation using whole-body water-fat mri. *Journal of Magnetic Resonance Imaging*, 41(6):1558–1569, 2015.
- [Keys *et al.*, 2014] Ancel Keys, Flaminio Fidanza, Martti J Karvonen, Noburu Kimura, and Henry L Taylor. Indices of relative weight and obesity. *International journal of epidemiology*, 43(3):655–665, 2014.
- [Kim *et al.*, 2022] Hyomin Kim, Hyeonseo Nam, Jungeon Kim, Jaesik Park, and Seungyong Lee. Laplacianfusion: Detailed 3d clothed-human body reconstruction. *ACM Transactions on Graphics (TOG)*, 41(6):1–14, 2022.
- [Kirillov *et al.*, 2023] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. Segment anything. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 4015–4026, 2023.
- [Küstner *et al.*, 2020] Thomas Küstner, Tobias Hepp, Marc Fischer, Martin Schwartz, Andreas Fritsche, Hans-Ulrich Häring, Konstantin Nikolaou, Fabian Bamberg, Bin Yang, Fritz Schick, et al. Fully automated and standardized segmentation of adipose tissue compartments via deep learning in 3d whole-body mri of epidemiologic cohort studies. *Radiology: Artificial Intelligence*, 2(6):e200010, 2020.
- [Lan *et al.*, 2022] Gongjin Lan, Yu Wu, Fei Hu, and Qi Hao. Vision-based human pose estimation via deep learning: A survey. *IEEE Transactions on Human-Machine Systems*, 53(1):253–268, 2022.
- [López *et al.*, 2012] Angel A López, Mey L Cespedes, Teófila Vicente, Matias Tomas, Miguel Bennasar-Veny, Pedro Tauler, and Antoni Aguilo. Body adiposity index utilization in a spanish mediterranean population: comparison with the body mass index. *PloS one*, 7(4):e35281, 2012.
- [Nana *et al.*, 2022] A Nana, JMD Staynor, S Arlai, A El-Sallam, N Dhungel, and MK Smith. Agreement of anthropometric and body composition measures predicted from 2d smartphone images and body impedance scales with criterion methods. *Obesity Research & Clinical Practice*, 16(1):37–43, 2022.
- [Nascimento *et al.*, 2025] Navar Medeiros M Nascimento, Pedro Cavalcante de Sousa Junior, Pedro Yuri Rodrigues Nunes, Suane Pires Pinheiro da Silva, Luiz Lannes Loureiro, Victor Zaban Bittencourt, Valden Luis Matos Capistrano Junior, and Pedro Pedrosa Rebouças Filho. New advances in body composition assessment with shapednet: A single image deep regression approach. *Pattern Recognition Letters*, 188:88–94, 2025.
- [Oord *et al.*, 2018] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.
- [Organization and others, 2021] World Health Organization et al. *Ethics and governance of artificial intelligence for health: WHO guidance: executive summary*. World Health Organization, 2021.
- [Radwan *et al.*, 2018] Hadia Radwan, Rami A Ballout, Hayder Hasan, Nader Lessan, Mirey Karavetian, and Rana Rizk. The epidemiology and economic burden of obesity and related cardiometabolic disorders in the united arab emirates: a systematic review and qualitative synthesis. *Journal of obesity*, 2018(1):2185942, 2018.



|     |                                                                   |                                                                                                                                                             |     |
|-----|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 669 | [Rezatofighi <i>et al.</i> , 2019]                                | Hamid Rezatofighi, Nathan Tsoi,                                                                                                                             | 725 |
| 670 |                                                                   | JunYoung Gwak, Amir Sadeghian, Ian Reid, and Silvio                                                                                                         | 726 |
| 671 |                                                                   | Savarese. Generalized intersection over union: A metric                                                                                                     | 727 |
| 672 |                                                                   | and a loss for bounding box regression. In <i>Proceedings of</i>                                                                                            |     |
| 673 |                                                                   | <i>the IEEE/CVF conference on computer vision and pattern</i>                                                                                               | 728 |
| 674 |                                                                   | <i>recognition</i> , pages 658–666, 2019.                                                                                                                   | 729 |
| 675 | [Robinette <i>et al.</i> , 1999]                                  | Kathleen M Robinette, Hans Daa-                                                                                                                             | 730 |
| 676 |                                                                   | nen, and Eric Paquet. The caesar project: a 3-d surface                                                                                                     | 731 |
| 677 |                                                                   | anthropometry survey. In <i>Second international conference</i>                                                                                             |     |
| 678 |                                                                   | <i>on 3-D digital imaging and modeling (cat. No. PR00062)</i> ,                                                                                             | 732 |
| 679 |                                                                   | pages 380–386. IEEE, 1999.                                                                                                                                  | 733 |
| 680 | [Ruiz <i>et al.</i> , 2022]                                       | Nataniel Ruiz, Miriam Bellver, Timo                                                                                                                         | 734 |
| 681 |                                                                   | Bolkart, Ambuj Arora, Ming C Lin, Javier Romero, and                                                                                                        | 735 |
| 682 |                                                                   | Raja Bala. Human body measurement estimation with ad-                                                                                                       | 736 |
| 683 |                                                                   | versarial augmentation. In <i>2022 International Conference</i>                                                                                             |     |
| 684 |                                                                   | <i>on 3D Vision (3DV)</i> , pages 219–230. IEEE, 2022.                                                                                                      | 737 |
| 685 | [Saxena and Mittal, 2022]                                         | Arpna Saxena and Sangeeta Mit-                                                                                                                              | 738 |
| 686 |                                                                   | tal. Internet of medical things (iomt) security and privacy:                                                                                                | 739 |
| 687 |                                                                   | A survey of recent advances and enabling technologies. In                                                                                                   | 740 |
| 688 |                                                                   | <i>Proceedings of the 2022 fourteenth international confer-</i>                                                                                             | 741 |
| 689 |                                                                   | <i>ence on contemporary computing</i> , pages 550–559, 2022.                                                                                                | 742 |
| 690 | [Sengupta <i>et al.</i> , 2021]                                   | Akash Sengupta, Ignas Budvytis,                                                                                                                             |     |
| 691 |                                                                   | and Roberto Cipolla. Probabilistic 3d human shape and                                                                                                       | 743 |
| 692 |                                                                   | pose estimation from multiple unconstrained images in the                                                                                                   | 744 |
| 693 |                                                                   | wild. In <i>Proceedings of the IEEE/CVF Conference on</i>                                                                                                   | 745 |
| 694 |                                                                   | <i>Computer Vision and Pattern Recognition</i> , pages 16094–                                                                                               | 746 |
| 695 |                                                                   | 16104, 2021.                                                                                                                                                | 747 |
| 696 | [Thomas <i>et al.</i> , 2013]                                     | E Louise Thomas, JA Fitzpatrick,                                                                                                                            | 748 |
| 697 |                                                                   | SJ Malik, Simon D Taylor-Robinson, and Jimmy D Bell.                                                                                                        | 749 |
| 698 |                                                                   | Whole body fat: content and distribution. <i>Progress in nu-</i>                                                                                            | 750 |
| 699 |                                                                   | <i>clear magnetic resonance spectroscopy</i> , 73:56–80, 2013.                                                                                              | 751 |
| 700 | [Velázquez-Alva <i>et al.</i> , 2022]                             | María Consuelo Velázquez-                                                                                                                                   | 752 |
| 701 |                                                                   | Alva, María Esther Irigoyen-Camacho, Marco Antonio                                                                                                          | 753 |
| 702 |                                                                   | Zepeda-Zepeda, Itzam Rangel-Castillo, Isabel Arrieta-                                                                                                       | 754 |
| 703 |                                                                   | Cruz, Luciano Mendoza-Garcés, Antonio Castaño-                                                                                                              | 755 |
| 704 |                                                                   | Seiquer, Javier Flores-Fraile, and Roger Gutiérrez-Juárez.                                                                                                  | 756 |
| 705 |                                                                   | Comparison of body fat percentage assessments by bio-                                                                                                       | 757 |
| 706 |                                                                   | electrical impedance analysis, anthropometrical prediction                                                                                                  | 758 |
| 707 |                                                                   | equations, and dual-energy x-ray absorptiometry in older                                                                                                    | 759 |
| 708 |                                                                   | women. <i>Frontiers in nutrition</i> , 9:978971, 2022.                                                                                                      | 760 |
| 709 | [World Health Organization, 2022]                                 | World Health                                                                                                                                                | 761 |
| 710 |                                                                   | Organization. Noncommunicable diseases                                                                                                                      | 762 |
| 711 |                                                                   | country profiles 2022: United arab emi-                                                                                                                     | 763 |
| 712 |                                                                   | rates. <a href="https://www.who.int/publications/m/item/noncommunicable-diseases-country-profiles-2022">https://www.who.int/publications/m/item/</a>        | 764 |
| 713 |                                                                   | <a href="https://www.who.int/publications/m/item/noncommunicable-diseases-country-profiles-2022">noncommunicable-diseases-country-profiles-2022</a> , 2022. | 765 |
| 714 |                                                                   | WHO Regional Office for the Eastern Mediterranean;                                                                                                          | 766 |
| 715 |                                                                   | UAE country profile; accessed January 2026.                                                                                                                 |     |
| 716 | [World Health Organization, 2024]                                 | World Health Organiza-                                                                                                                                      | 767 |
| 717 |                                                                   | tion. Body mass index (bmi), 2024. Accessed January                                                                                                         | 768 |
| 718 |                                                                   | 2026.                                                                                                                                                       | 769 |
| 719 | [World Obesity Federation, 2023]                                  | World Obesity Federa-                                                                                                                                       | 770 |
| 720 |                                                                   | tion. World obesity atlas 2023: United arab emirates                                                                                                        | 771 |
| 721 |                                                                   | country profile. <a href="https://data.worldobesity.org/country/united-arab-emirates-225/">https://data.worldobesity.org/country/</a>                       | 772 |
| 722 |                                                                   | <a href="https://data.worldobesity.org/country/united-arab-emirates-225/">united-arab-emirates-225/</a> , 2023. Accessed January 2026.                      | 773 |
| 723 | [Xie <i>et al.</i> , 2015]                                        | Bowen Xie, Jesus I Avila, Bennett K Ng,                                                                                                                     | 774 |
| 724 |                                                                   | Bo Fan, Victoria Loo, Vicente Gilsanz, Thomas Hangart-                                                                                                      | 775 |
|     |                                                                   | ner, Heidi J Kalkwarf, Joan Lappe, Sharon Oberfield, et al.                                                                                                 | 776 |
|     |                                                                   | Accurate body composition measures from whole-body                                                                                                          | 777 |
|     |                                                                   | silhouettes. <i>Medical physics</i> , 42(8):4668–4677, 2015.                                                                                                | 778 |
|     | [Yan <i>et al.</i> , 2021]                                        | Song Yan, Johan Wirta, and Joni-Kristian                                                                                                                    | 779 |
|     |                                                                   | Kämäräinen. Silhouette body measurement benchmarks.                                                                                                         | 780 |
|     |                                                                   | In <i>2020 25th international conference on pattern recogni-</i>                                                                                            |     |
|     |                                                                   | <i>tion (ICPR)</i> , pages 7804–7809. IEEE, 2021.                                                                                                           |     |
|     | [Zhu <i>et al.</i> , 2023]                                        | Haidong Zhu, Zhaoheng Zheng, and Ram                                                                                                                        |     |
|     |                                                                   | Nevatia. Gait recognition using 3-d human body shape in-                                                                                                    |     |
|     |                                                                   | ference. In <i>Proceedings of the IEEE/CVF Winter Confer-</i>                                                                                               |     |
|     |                                                                   | <i>ence on Applications of Computer Vision</i> , pages 909–918,                                                                                             |     |
|     |                                                                   | 2023.                                                                                                                                                       |     |
|     | [Zolfaghari <i>et al.</i> , 2021]                                 | Mohammadreza Zolfaghari,                                                                                                                                    |     |
|     |                                                                   | Yi Zhu, Peter Gehler, and Thomas Brox. Crossclr:                                                                                                            |     |
|     |                                                                   | Cross-modal contrastive learning for multi-modal video                                                                                                      |     |
|     |                                                                   | representations. In <i>Proceedings of the IEEE/CVF interna-</i>                                                                                             |     |
|     |                                                                   | <i>tional conference on computer vision</i> , pages 1450–1459,                                                                                              |     |
|     |                                                                   | 2021.                                                                                                                                                       |     |
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